Coursework Title:	Research Report
Course Name:	Human-Computer Interaction
Course Code:	DECO2500/7250
Weighting:	15%
Hand-in Date:	26 August 2022 by 4:00 PM (AEST)
Hand-in Method:	Blackboard

1. Introduction

This is an **individual** piece of work. You are not allowed to work with anyone else during the completion of this assignment.

You must submit your work on Blackboard on or before the due date/time.

You have just been employed by a technology development company into their Research and Development team. They are exploring future opportunities for technology development in a number of domains. The intention is to form a number of teams to undertake explorations of what sorts of technology might be useful for people and to develop prototypes that will act as illustrations of the type of technology, as well as how that technology should be designed so that it has the greatest chance of people buying that system. However, before that can occur, they first need you to undertake some initial research into the problem.

In this stage of the development process, you are *individually* undertaking your first cycle through the interaction design process. The key focus of this cycle is to identify the issues and opportunities: what are the issues and what do people want in terms of technology within this domain? Critical is the need to ensure that the technology will be considered valuable and useful by people. To this end, it is important to understand people's needs and the issues that they are facing both with what the proposed technology is meant to deliver and the problems that they experience with current technology. The deliverable of this assessment task will be a report that captures your findings from your first cycle through the interaction design process.

2. Learning Objective Being Assessed

2 Relate cognitive, social, behavioural, and emotional theories to the design of effective human-system interaction.

3 Use a range of design research methods, which involve the people from target user groups – such as ethnography, contextual inquiry, interviews, and surveys – in the process of interactive system design.

4 Apply iterative interaction design techniques, such as scenario development, prototyping, and participatory design in the design of an interactive system.

6 Apply HCI knowledge, in terms of system accessibility, within appropriate user-interface design for a broad range of user groups.

3. Details of the Task

This task includes first individually performing a contextual inquiry, which will inform the group aspects of the course. The aim of this task is to illustrate your initial research and design ideas around a central subject. Your report should illustrate a clear reflection of your contextual inquiry and should clearly exhibit the initial research that has been undertaken.

In this task, you are required to identify an issue within a particular domain that you would like to explore further, gather information through research, establish your initial requirements and sketch out potential/initial solutions. Potential domains that you can explore include:

- *Behaviour Change:* Developing apps that help people change their habits is a growing domain and presents some unique opportunities (and challenges) in HCI. E.g. people may want to reduce food waste, decrease their consumption, save money, or stop procrastinating. Whatever it is, when it comes to persuasive technology, there is a range of HCI considerations that support the development of a successful app in this domain.

Image source: https://www.idownloadblog.com/2014/01/22/optimizeme-app/

- *Discovering:* People like to explore and learn about our world. It may be to learn more about their own neighbourhood, explore hidden features of a city, discover local wildlife, find out what is going on or to learn more about themselves. As people get out into the world there are extra considerations when developing apps. For example: if people are walking around a city with their attention on your app, could they inadvertently walk in front of a moving vehicle?

Image source: https://blog.nhbs.com/how-to-guides/ten-favourite-free-apps-wildlife-lovers/

- *Transacting:* What we used to do face-to-face we are now increasingly doing through interactive systems. We buy, sell, bank, enrol, vote, converse, share, book, cancel, insure, pay, chat, enquire, play, help, socialise, support and so much more through computer mediated systems. And the way we interact with these systems is continually changing. How do we design these interactions so that people engage with them, can make sense of them, and use them effectively?

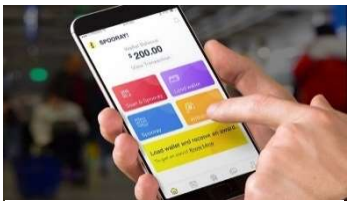

Image source: https://www.peerbits.com/blog/develop-a-peer-to-peer-payment-app.html

- *Health and Wellbeing*: Technology in within the health domain is a huge business that allows users and health professionals to continuously track and manage certain health data, perform disease surveillance, deliver treatment support, track epidemic outbreaks and manage chronic disease. Such applications can also collect community and clinical health data, deliver healthcare information to practitioners, researchers, and patients, provide real-time monitoring of patient vital signs, as well as training and collaboration of health workers.

Image source: https://www.lakeforestmba.edu/blog/impact-mhealth-technology-american-healthcare/

The focus is to start identifying what people might want. You need to understand the "problem" that your app or system may seek to address and people's experience of that problem. Then you will need to start creating an idea for that app or system and create a conceptual design for that application including some of your ideas for functionality and how people will interact with it. This will all need to be captured within your research report.

3.1 Research Report Deliverable

Contextual Inquiry

Within your research report, you are required to present your ideas within the following four areas:

a) **Research: Selection of domain and issue(s) that have been identified:** As you are starting from scratch (i.e., you don't have an existing system that you are building on), the

first step is to select a domain and identify an issue, which will emerge after conducting a broad range of research into your domain.

- Multiple activities should be undertaken at this stage, including (but not limited to) – researching existing applications and systems, conducting surveys and interviewing/observing people.

1) **b) Conceptual Design: Initial requirements:** Once you have identified an issue, you need to establish a clear conceptual design of your app or system that is linked to the research you have undertaken. Within your report you should clearly demonstrate:

a) - System concept, which can include (but is not limited to):
  i) - Problem statement
    - High level description of how the system works (i) o
    - Interaction paradigm
    - Interaction mode
    - Metaphors – system, interaction & interface
- Initial Design Guidelines
- Initial System Requirements

c) Low Fidelity Prototype: Initial sketches of possible solutions: The purpose of your prototype is to enable you to develop an understanding of the mental model your intended user applies to your app or system and gather further information from people about the problem that your application is seeking to address

- The prototype must encompass sufficient details of your proposed application to support these aims. You will need to include in your report an explanation of your prototype, in particular your rationale for the design of the prototype (you should refer back to your conceptual design and the data that you gathered during the contextual inquiry stages). You will need to include photographs of your paper prototype in your report. It is important to remember that at this stage, your initial requirements and low fidelity prototype are not a fully developed system but rather the starting point.

d) Future Directions: Future design of the system: Identify future steps for both the design of your prototype/system, and for your process. For the future design of the system, you can identify things like:

- Possible additional features that could make sense for your user group or context
- Possible integration with external services or data that would make sense for your user group or context
- Anticipated difficulties users may encounter interacting with/making sense of your prototype

For your future process, you can identify things like:

- Particularly significant aspects or features of your prototype that will be important to evaluate
- Further activities—research, design, implementation, evaluation—that will need to be conducted to improve your conceptual design

3.2 Report Guidelines

The main body of your report should not exceed 8 pages (excluding title page, contents, references, and appendix). The report should be suitably formatted, structured logically and be free from grammar and spelling errors. To prepare your report, you are advised to use a word processor (such as Microsoft Word or OpenOffice/LibreOffice) that supports spelling and grammar checking.

You are also required to appropriately reference all resources and personal communication in your report and include a reference list. IEEE referencing style is recommended but not required. It is your choice what referencing style you ultimately use. Whichever style you choose, ensure you are consistent throughout your document.

4. What You Should Hand In

You should submit a **single electronic file** on Blackboard on or before the due date/time.

Your submission must be a single word-processed document that contains your report (*Microsoft Word (.doc/.docx), OpenOffice Text (.odt)* or *Adobe PDF* formats only).

No other forms of submission are allowed without prior written permission from the course coordinator.

5. Grading Scheme/Assessment Criteria*

Assessment Criteria		Proportion weighting for criteria (%)
Report	Research	25
	Conceptual Design	25
	Low Fidelity Prototype	20
	Future Directions	20
	Professionalism of Report	10

**please refer to the rubric on page 7/8 for a detailed breakdown of each criteria*

6. Guidelines

- Correctly reference any resources that you use.

7. Recommended Resources

Making use of these specific tools and resources may be helpful for completing the task:

- Lecture/Studio Materials
- The Internet
- Course-specific textbooks, such as:
  - Chapter 11, Preece, J., Rogers, Y., & Sharp, H. (2015). *Interaction design beyond human-computer interaction* (Fifth ed.)
- UQ Resources, such as:
  - https://my.uq.edu.au/information-and-services/student-support/study-skills/assignment-writing/assignment-types
  - https://my.uq.edu.au/information-and-services/student-support/study-skills/assignment-writing
  - https://my.uq.edu.au/information-and-services/student-support/study-skills
  - https://web.library.uq.edu.au/research-tools-techniques/google-scholar
  - https://guides.library.uq.edu.au/referencing

8. Late Submission / Extenuating Circumstances

Please refer to the ECP for full details about the University's policies and guidelines regarding assessment.

9. Academic Misconduct

The University defines Academic Misconduct as involving 'a range of unethical behaviours that are designed to give a student an unfair and unearned advantage over their peers'.

The University takes Academic Misconduct very seriously and any suspected cases will be investigated through the University's standard policy. If you are found guilty, you may be expelled from the University with no award.

It is the responsibility of the student to ensure that you understand what constitutes Academic Misconduct and to ensure that you do not break the rules. If you are unclear about what is required, please ask.

It is also the responsibility of the student to take reasonable precautions to guard against unauthorised access by others to their work, however stored in whatever format, both before and after assessment.

For more information, you are directed to the following University web pages:

- Information regarding ***ITEE Student Misconduct (Including Plagiarism)***
  - https://www.itee.uq.edu.au/itee-student-misconduct-including-plagiarism
- Information regarding ***Academic Integrity and Misconduct***:
  - https://my.uq.edu.au/information-and-services/manage-my-program/student-integrity-and-conduct/academic-integrity-and-student-conduct
  - http://ppl.app.uq.edu.au/content/3.60.04-student-integrity-and-misconduct
- Information on ***Student Services:***
  - https://www.uq.edu.au/student-services/
- Information regarding ***Referencing***:
  - https://web.library.uq.edu.au/research-tools-techniques/digital-essentials/write-cite-submit?p=2#2
  - https://web.library.uq.edu.au/research-tools-techniques/referencing-style-guides
- Information regarding the ***Student Charter***:
  - https://ppl.app.uq.edu.au/content/3.60.01-student-charter
- Information regarding ***Plagiarism***:
  - https://web.library.uq.edu.au/node/4221/1#1

10. Research Report Grading Criteria

Criteria/ Weighting	High Distinction *Demonstrated evidence of **exceptional** achievement in relation to assessment criteria*	Distinction *Demonstrated evidence of **advanced** achievement in relation to assessment criteria*	Credit *Demonstrated evidence of **proficient** achievement in relation to assessment criteria*	Pass *Demonstrated evidence of **functional** achievement in relation to assessment criteria*	Fail ***Minimal to Absence** of evidence of achievement in relation to assessment criteria*	X (None) *No assessable work has been received*
Research (25%)	Research has demonstrated **exceptional** breadth and illustrates mastery of the content in the area of HCI. Your application of this research captures **superior** detail to enable a **thorough** understanding of people and context.	Research has demonstrated an **advanced** level of breadth and illustrates **substantial** knowledge of fundamental concepts in the area of HCI. Your application of this research captures a **suitable** level of detail to enable a **reasonable** understanding of people and context.	Research has demonstrated a **competent** level of breadth and illustrates **good** knowledge of fundamental concepts in the area of HCI. Your application of this research captures **adequate** detail to enable a **satisfactory** understanding of people and context.	Research has demonstrated an **acceptable** level of breadth and illustrates **adequate** knowledge of fundamental concepts in the area of HCI. Your application of this research captures **minimal** detail to enable a **limited** understanding of people and context.	Research has demonstrated a **severe lack** of detail and illustrates **deficiencies** or **no demonstratable evidence** of understanding of the fundamental concepts in the area of HCI.	This component is missing
Conceptual Design (25%)	Conceptual design is **clearly** linked to the research undertaken. It is **highly detailed** and demonstrates **creative** application of knowledge and skills. The system concept, guidelines and requirements have **all** been presented to an **exceptional** standard.	Conceptual design is **mostly** linked to the research undertaken. It is **very detailed** and demonstrates **extensive** application of knowledge and skills. The system concept, guidelines and requirements have **mostly** been presented to a **very high** standard.	Conceptual design is **largely** linked to the research undertaken. It is **suitably detailed** and demonstrates **effective** application of knowledge and skills. The system concept, guidelines and requirements have **largely** been presented to a **suitable** standard.	Conceptual design is **barely** linked to the research undertaken. It is **satisfactorily detailed** and demonstrates **acceptable** application of knowledge and skills. The system concept, guidelines and requirements have **barely** been presented to an **acceptable** standard.	Conceptual design **is not** linked to the research undertaken. It is **severely lacking detail** and demonstrates an **inability** to apply knowledge and skills. The system concept, guidelines and requirements **do not sufficiently** meet the requirements.	

Criteria/ Weighting	High Distinction *Demonstrated evidence of **exceptional** achievement in relation to assessment criteria*	Distinction *Demonstrated evidence of **advanced** achievement in relation to assessment criteria*	Credit *Demonstrated evidence of **proficient** achievement in relation to assessment criteria*	Pass *Demonstrated evidence of **functional** achievement in relation to assessment criteria*	Fail ***Minimal to Absence** of evidence of achievement in relation to assessment criteria*	X (None) *No assessable work has been received*
Low Fidelity Prototype (20%)	Prototype is **clearly** linked to the research undertaken and conceptual design. It provides an **exceptional** representation of the design, including details of appearance and interaction behaviour. Demonstrates **creative** application of knowledge and skills.	Prototype is **mostly** linked to the research undertaken and conceptual design. It provides an **advanced** representation of the design and **mostly** conveys aspects of the appearance and/or interaction behaviour. Demonstrates **extensive** application of knowledge and skills.	Prototype is **largely** linked to the research undertaken and conceptual design. It provides a **competent** representation of the design and **generally** conveys aspects of the appearance and/or interaction behaviour. Demonstrates **effective** application of knowledge and skills.	Prototype is **barely** linked to the research undertaken and conceptual design. It provides an **acceptable** representation of the design. Conveys a **limited** representation of aspects of the appearance and/or interaction behaviour. Demonstrates **acceptable** application of knowledge and skills.	Prototype **is not** linked to the research undertaken and conceptual design. It is **severely lacking** detail. As such, the representation of the design does not effectively convey aspects of either the appearance or interaction behaviour. Demonstrates an **inability** to apply knowledge and skills.	This component is missing
Future Directions (20%)	Future directions are **highly detailed** and demonstrates **creative** application of knowledge and skills. **Both** future design and process have been presented to an **exceptional** standard.	Future directions are **very detailed** and demonstrates **extensive** application of knowledge and skills. **Both** future design and process have **mostly** been presented to a **very high** standard.	Future directions are **suitably detailed** and demonstrates **effective** application of knowledge and skills. Future design and/or process have **largely** been presented to a **suitable** standard.	Future directions are **satisfactorily detailed** and demonstrates **acceptable** application of knowledge and skills. Future design and/or process have **barely** been presented to an **acceptable** standard.	Future directions are **severely lacking detail** and demonstrates an **inability** to apply knowledge and skills and **does not sufficiently** meet the requirements.	
Professionalism of Report (10%)	The report has been presented to an **exceptional** standard. It has been organised **very well** and in a logical manner and is **highly** easy to follow. It communicates at an **expert** level with minimal to no spelling/grammar errors. The page limit has been adhered to **well**.	The report has been presented to an **advanced** standard. It has been organised **quite well** and is **very** easy to follow. It communicates at a **professional** level. May contain limited spelling/ grammar errors. The page limit has been **reasonably** adhered to.	The report has been presented to a **competent** standard. It has been organised **well** and is **quite** easy to follow. It communicates at an **effective** level. May contain some spelling/ grammar errors. The page limit has been **mostly** adhered to.	The report has been presented to an **acceptable** standard. It has been organised **adequately** but may be hard to follow. It communicates at an **appropriate** level. May contain numerous spelling/grammar errors. The page limit **has not** been adhered to well.	The report **fails** to meet the standard at this level. It is **disorganised** and communicates information or ideas in ways that are frequently incomplete, confusing, and not appropriate. Written expression is **disjointed,** and the report is badly structured with constant errors in spelling/grammar.	